

Introductory Statistics for Econometrics

Unit 7. Law of Large Numbers and Central Limit Theorem

Random samples, Chebyshev's inequality, the Law of Large Numbers,
convergence in distribution, and the Central Limit Theorem

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Learning outcomes. After this unit, the student should be able to (i) state the i.i.d. random sample setup and the properties of the sample mean; (ii) use Chebyshev's inequality to bound deviation probabilities; (iii) state and apply the (weak) Law of Large Numbers; (iv) distinguish convergence in probability from convergence in distribution; (v) state and apply the Central Limit Theorem; (vi) explain why both LLN and CLT can hold simultaneously.

Source material: condensed from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 15 and the introduction of Lecture 17.

1 Random samples and the sample mean

Until now we computed probabilities and moments *given* a known distribution. From this unit onward the situation is reversed: the distribution is unknown and we observe a random sample drawn from it which is the usual thing in econometrics.

Definition 1.1 (Random sample). A *random sample* of size n is a sequence $X_1, \dots, X_n$ of random variables that are *independent and identically distributed* (i.i.d.) with common density f_X .

Definition 1.2 (Sample mean).

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Proposition 1.3 (Mean and variance of the sample mean). If $X_1, \dots, X_n$ are i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$,

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

If in addition the X_i are normal, $\bar{X}_n \sim \mathcal{N}(\mu, \sigma^2/n)$.

The fact that $\text{Var}(\bar{X}_n) \rightarrow 0$ as $n \rightarrow \infty$ is the seed of the Law of Large Numbers below.

2 Chebyshev's inequality

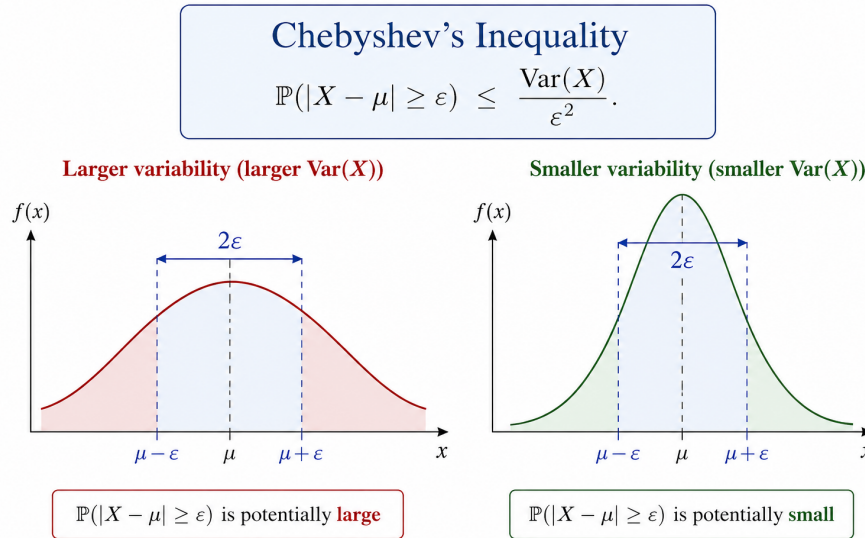
Chebyshev's inequality is the workhorse tool that turns a finite variance into a probabilistic bound on deviations from the mean.

Theorem 2.1 (Chebyshev's inequality). Let X be a random variable with $\mathbb{E}[X] = \mu$ and $\text{Var}(X) < \infty$. Then for every $\varepsilon > 0$,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\text{Var}(X)}{\varepsilon^2}.$$

Sketch. $\text{Var}(X) = \int (t - \mu)^2 f_X(t) dt \geq \int_{|t - \mu| \geq \varepsilon} (t - \mu)^2 f_X(t) dt \geq \varepsilon^2 \mathbb{P}(|X - \mu| \geq \varepsilon).$ $\square$

Chebyshev makes the slogan “variance measures dispersion” literal: it bounds the probability mass beyond any chosen distance from the mean. In other words this inequality measures how far is our random variable away its true mean. The smaller the variability in the data the more concentrated the random variable will be to the true mean.



Example 2.2 (Bounded deviation of the sample mean). By Proposition 1.3, $\text{Var}(\bar{X}_n) = \sigma^2/n$. Applied to $\bar{X}_n$,

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \xrightarrow{n \rightarrow \infty} 0.$$

This is essentially the LLN below.

3 Convergence in probability and the Law of Large Numbers

Definition 3.1 (Convergence in probability). A sequence of random variables $\{Y_n\}$ *converges in probability* to a constant c , written $Y_n \xrightarrow{p} c$, if for every $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P}(|Y_n - c| > \varepsilon) = 0.$$

Theorem 3.2 (Weak Law of Large Numbers). If $X_1, \dots, X_n$ are i.i.d. with $\mathbb{E}[X_i] = \mu$ and finite variance,

$$\bar{X}_n \xrightarrow{p} \mu \quad \text{as } n \rightarrow \infty.$$

Proof. Apply Chebyshev to $\bar{X}_n$: $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \sigma^2/(n\varepsilon^2) \rightarrow 0.$ $\square$

Reading. For large samples, the sample mean is, with very high probability, arbitrarily close to the population mean. “Sample averages estimate population means.”

Example 3.3 (Standardisation of measurement units). A medieval ruler measuring “one foot” from a single person’s foot is noisy. Averaging the foot lengths of 16 randomly chosen people reduces the variance of the unit by $1/16$, in line with $\text{Var}(\bar{X}_n) = \sigma^2/n$. This is one of the earliest documented applications of the LLN — averaging averages out idiosyncratic noise.

Example 3.4 (The wisdom of crowds). n voters choose between two candidates. A fraction 2ε vote correctly with certainty (informed); the remaining $1 - 2\varepsilon$ vote by coin flip (uninformed). Define $X_i = 1$ if voter i chooses A. Then $\mathbb{E}[\bar{X}_n] = \frac{1}{2} + \varepsilon$ and $\text{Var}(\bar{X}_n) = (1 - 2\varepsilon)/(4n)$. Chebyshev gives

$$\mathbb{P}(\bar{X}_n < \tfrac{1}{2}) \leq \frac{1 - 2\varepsilon}{4n\varepsilon^2}.$$

For $2\varepsilon = 5\%$, $n \geq 95$ already suffices to keep the probability of electing the wrong candidate below 5% — even though 95% of voters are uninformed. Independence of the noise is what makes the LLN do its job.

Remark 3.5 (Independence is essential). Suppose a televised event sways the entire uninformed electorate in a common direction. Now the conditional variance of $\bar{X}_n$ given the event is small, but $\text{Var}(\mathbb{E}[\bar{X}_n \mid \text{event}])$ does *not* shrink with n . By the law of total variance (Unit 6), $\text{Var}(\bar{X}_n)$ stays bounded away from zero — and the LLN can fail. The lesson is general: *common shocks invalidate the LLN*, which is why economists worry so much about cross-sectional dependence and clustering of standard errors.

4 Convergence in distribution and the Central Limit Theorem

The LLN says that the sample mean concentrates on a point. To get a non-degenerate limit, we have to rescale by exactly the right amount.

Definition 4.1 (Convergence in distribution). A sequence of random variables $\{Y_n\}$ *converges in distribution* to a random variable Y with c.d.f. F_Y , written $Y_n \xrightarrow{d} Y$, if at every continuity point y of F_Y

$$\lim_{n \rightarrow \infty} \mathbb{P}(Y_n \leq y) = F_Y(y).$$

Convergence in distribution is about the *shape* of the distribution. It is weaker than convergence in probability: $Y_n \xrightarrow{p} c$ implies $Y_n \xrightarrow{d} c$, but not the reverse.

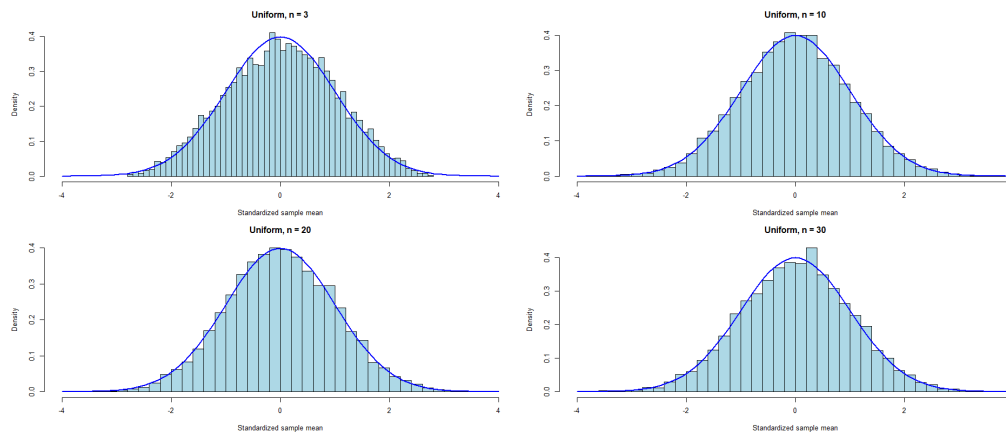
Theorem 4.2 (Central Limit Theorem). Let $X_1, \dots, X_n$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 \in (0, \infty)$. Then

$$\sqrt{n} \frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1), \quad \text{equivalently} \quad \sqrt{n} (\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

Why the normal limit? The sum of two independent normals is again normal, so the normal family is *stable* under independent addition. The only stable family with finite variance is normal. The CLT identifies the normal as the unique attractor of the rescaled sample mean.

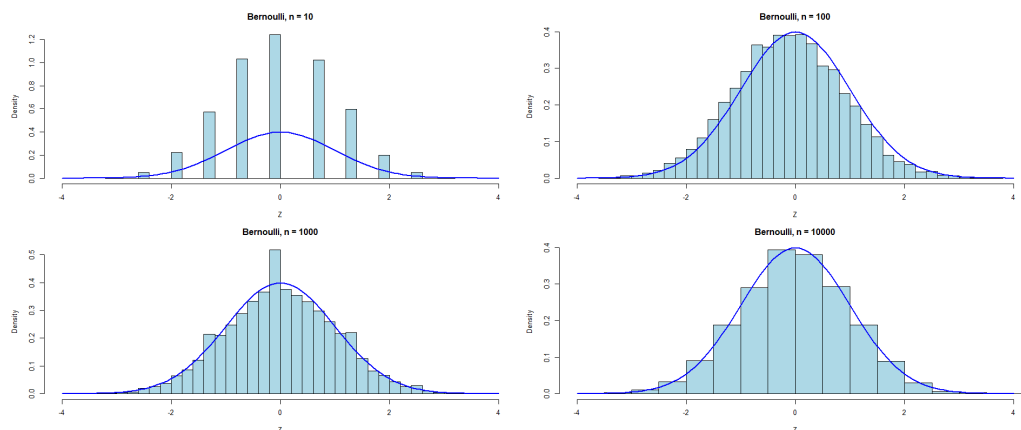
Example 4.3 (CLT in action with uniforms). $X_i \stackrel{\text{iid}}{\sim} U[0, 1]$, $\mu = 1/2$, $\sigma^2 = 1/12$. The standardised sum $Z_n = \sqrt{n}(\bar{X}_n - 1/2)/\sqrt{1/12}$ has p.d.f. that converges to φ . Numerically, the approximation is excellent for $n \geq 30$; for $n = 3$ already the standardised triangle-on-triangle convolution is visually indistinguishable from φ in the bulk.

Below we present some simulations to show the power of the CLT. We generate a uniform random variable as defined in the example and compute Z_n 1000 times. We then plot the resulting histogram together with the standard normal density, shown in dark blue. As the sample size increases, the histogram becomes closer to the shape of the Gaussian density. This illustrates the convergence in distribution of Z_n to a standard normal random variable.

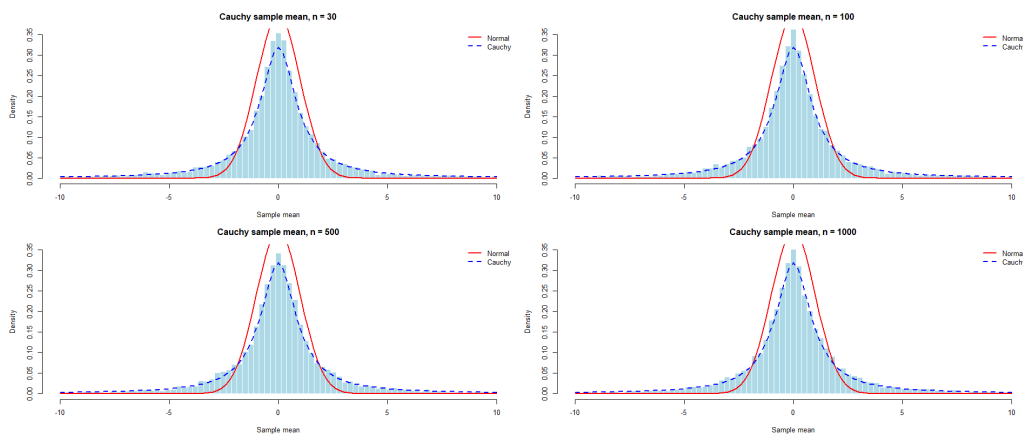


Example 4.4 (Binomial revisited). DeMoivre–Laplace (Unit 7) is the special case $X_i \sim \text{Bernoulli}(p)$. The sample mean $\bar{X}_n = Y_n/n$ where $Y_n \sim B(n, p)$, and $\sqrt{n}(\bar{X}_n - p) \xrightarrow{d} \mathcal{N}(0, p(1-p))$. Computing binomial tail probabilities via the normal table is a CLT calculation in disguise.

Below we present some simulations to show the power of the CLT. We generate a n bernoulli random variables as defined in the example with $p = 0.4$ and compute $Z_n = \sqrt{n}(\bar{X}_n - 0.4)/(0.24)$ 1000 times. We then plot the resulting histogram together with the standard normal density, shown in dark blue. As the sample size increases, the histogram becomes closer to the shape of the Gaussian density. This illustrates the convergence in distribution of Z_n to a standard normal random variable.



Remark 4.5 (Practical rules of thumb). **The CLT does *not* generally hold without finite variance.** A famous counterexample is the Cauchy distribution (check the image below). The reason is that the normal approximation relies on the idea that the normalized sum is formed by many small contributions, none of which dominates the total. When the variance is infinite, extreme observations may be too influential, and a single observation can have a non-negligible effect on the whole sum. In that case, the usual $\sqrt{n}$ -normalization need not produce a Gaussian limit.



5 Why this is the foundation of econometrics

Almost every estimator we will compute in Units 9–10 satisfies a “CLT-type” result of the form

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, V_\theta),$$

where V_θ is some asymptotic variance. Three immediate consequences:

1. Estimators are *consistent*: $\hat{\theta}_n \xrightarrow{p} \theta_0$ (Unit 9). LLN feeds consistency.
2. Standard errors and confidence intervals can be built using the normal table (Unit 10). CLT feeds inference.
3. Testing reduces to comparing the standardised statistic to a $\mathcal{N}(0, 1)$, t , χ^2 or F critical value.

The OLS estimator $\hat{\beta} = (X^\top X)^{-1} X^\top \mathbf{Y}$ is the leading example: under some regularity conditions it is asymptotically normal with covariance $\sigma^2(X^\top X)^{-1}$ — a multivariate CLT statement.

Summary

- *Random sample*: i.i.d. $X_1, \dots, X_n \sim f_X$.
- *Sample mean*: $\bar{X}_n$ has mean μ and variance σ^2/n .
- *Chebyshev*: $\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \text{Var}(X)/\varepsilon^2$.
- *LLN*: $\bar{X}_n \xrightarrow{p} \mu$.
- *CLT*: $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$.
- LLN and CLT coexist: the sample mean converges to μ at rate $1/\sqrt{n}$, and the residual is asymptotically normal.

Suggested exercises

1. Use Chebyshev’s inequality to find the smallest n such that $\mathbb{P}(|\bar{X}_n - \mu| > 0.1) \leq 0.05$ when $\sigma^2 = 4$. Compare with the (sharper) bound from the CLT.
2. Let $X_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(0.4)$ and $\bar{X}_n$ the sample mean for $n = 100$. Use the CLT to estimate $\mathbb{P}(\bar{X}_n \geq 0.5)$.
3. Consider $X_i \sim \text{Exp}(1)$. Find the asymptotic distribution of $\sqrt{n}(\bar{X}_n - 1)$ and compare with simulated histograms for $n = 10, 50, 200$.

4. (Wisdom of crowds.) In Example 3.4, suppose the uninformed voters are not independent but split into K equally sized blocs that vote together. Show that $\text{Var}(\bar{X}_n)$ does not vanish as $n \rightarrow \infty$ unless $K \rightarrow \infty$.
5. Show that if $Y_n \xrightarrow{d} c$ for a constant c , then $Y_n \xrightarrow{p} c$. (The two notions agree on degenerate limits.)
6. (CLT failure.) Let $X_i \stackrel{\text{iid}}{\sim} \text{Cauchy}$. Show that $\bar{X}_n$ is itself Cauchy for every n , so neither LLN nor CLT can hold.